STAT 339 METHODS OF LINEAR ALGEBRA
Linear Transformations

Edward Acheampong, PhD

(Department of Statistics & Actuarial Science)

December 12, 2023

UNIVERSITY OF GHANA

Outline

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Introduction to Linear Transformations

Mapping

A function T that maps a vector space V into a vector space W

$$T : V \xrightarrow{mapping} W, \quad V, W \text{ are vector spaces,}$$

where V is the domain of T and W is the codomain of T

Image of $\mathbf{v}$ under T

If $\mathbf{v}$ is a vector space V and $\mathbf{w}$ is a vector in W such that

$$T(\mathbf{v}) = \mathbf{w},$$

then $\mathbf{w}$ is called the image of $\mathbf{v}$ under T
(For each $\mathbf{v}$, there is only $\mathbf{W}$)

Range of T

The set of all images of vectors in V

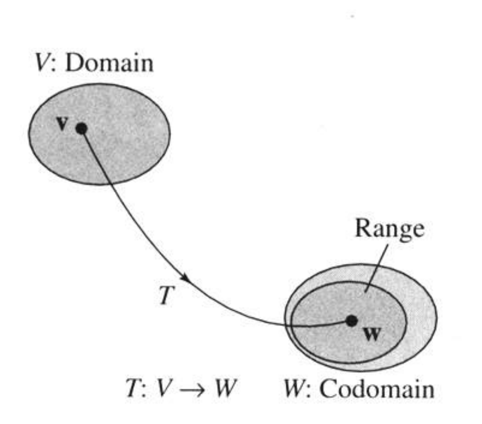
The Preimage of $\mathbf{w}$

The set of all $\mathbf{v}$ in V such that

$$T(\mathbf{v}) = \mathbf{w},$$

(For each $\mathbf{w}$, $\mathbf{v}$ may not be unique

The graphical representations of the domain, codomain, and rangee



Example

- For example, V is R^3, W is R^3, and T is the orthogonal projection of any vector (x, y, z) onto the xy-plane, i.e. $T(x, y, z) = (x, y, 0)$
  (we will use the above example many times to explain abstract notions)
- Then the domain is R^3, the codomain is R^3, and the range is xy-plane (a subspace of the codomian R^3)
- $(2, 1, 0)$ is the image of $(2, 1, 3)$
- The preimage of $(2, 1, 0)$ is $(2, 1, s)$, where s is any real number

Example (A function from R^2 into R^2)

$$T : R^2 \longrightarrow R^2$$

where $\mathbf{v} = (v_1, v_2) \in R^2$ and $T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$.
Find (a) the image of $\mathbf{v} = (-1, 2)$ (b) Find the preimage of $\mathbf{v} = (-1, 11)$

Solution

(a) $\mathbf{v} = (-1, 2)$

$\implies T(\mathbf{v}) = T(-1, 2) = (-1 - 2, -1 + 2(2)) = (-3, 3)$

(b) $T(\mathbf{v}) = \mathbf{w} = (-1, 11)$

$\implies T(\mathbf{v}) = T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2) = (-1, 11)$

$$\implies \begin{array}{r} v_1 - v_2 = -1 \\ v_1 + 2v_2 = 11 \end{array}$$

$\implies v_1 = 3, v_2 = 4.$

Thus, $\{(3, 4)\}$ *is the preimage of* $\mathbf{w} = (-1, 11)$.

Linear Transformation

V, W : vector spaces

$T : V \to W$: A linear transformation of V into W if the following two properties are true

1. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \forall \mathbf{u}, \mathbf{v} \in V$
2. $T(c\mathbf{u}) = cT(\mathbf{u}), \forall \mathbf{u}, \mathbf{v} \in R$

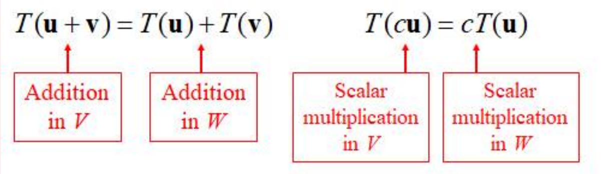
Introduction to Linear Transformations

Note

V, W : vector spaces

$T : V \to W$: A linear transformation of V into W if the following two properties are true

1. A linear transformation is said to be operation preserving (because the same result occurs whether the operations of addition and scalar multiplication are performed before or after the linear transformation is applied)



2. A linear transformation $T : V \to V$ from a vector space into itself is called a linear operator.

Example (Verifying a linear transformation T from R^2 into R^2)

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

Solution

Let $\mathbf{u} = (u_1, u_2), \mathbf{v} = (v_1, v_2)$ *a vectors in* R^2 *and* c *is any real number*

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

(1) *Vector addition*

$$\begin{aligned}
\mathbf{u} + \mathbf{v} =& (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) \\
T(\mathbf{u} + \mathbf{v}) =& T(u_1 + v_1, u_2 + v_2) \\
=& ((u_1 + v_1) - (u_2 + v_2), (u_1 + v_1) + 2(u_2 + v_2) \\
=& ((u_1 - u_2) + (v_1 - v_2), (u_1 + 2u_2) + (v_1 + 2v_2)) \\
=& (u_1 - u_2, u_1 + 2u_2) + (v_1 - v_2, v_1 + 2v_2) \\
=& T(\mathbf{u}) + T(\mathbf{v})
\end{aligned}$$

Solution

Let $\mathbf{u} = (u_1, u_2), \mathbf{v} = (v_1, v_2)$ *a vectors in* R^2 *and* c *is any real number*

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

(2) *Scalar multiplication*

$$\begin{aligned}
c\mathbf{u} &= c(u_1, u_2) = (cu_1, cu_2) \\
T(c\mathbf{u}) &= T(cu_1, cu_2) \\
&= (cu_1 - cu_2, cu_1 + 2cu_2) \\
&= c(u_1 - u_2, u_1 + 2u_2) \\
&= cT(\mathbf{u})
\end{aligned}$$

Therefore, T *is a linear transformation*

Example (Functions that are not linear transformations)

(a) $f(x) = \sin x$

$$\sin(x_1, x_2) \neq \sin(x_1) + \sin(x_2)$$
$$\sin(\frac{\pi}{2}, \frac{\pi}{3}) \neq \sin(\frac{\pi}{2}) + \sin(\frac{\pi}{3}$$

(b) $f(x) = x^2$

$$(x_1^2, x_2^2)^2 \neq x_1^2 + x_2^2$$
$$(1, 2)^2 \neq 1^2 + 2^2$$

(b) $f(x) = x + 1$

$$f(x_1 + x_2) \neq f(x_1) + f(x_2)$$
$$x_1 + x_2 + 1 \neq x_1 + x_2 + 2$$

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Introduction to Linear Transformations

Note: Two uses of the term "linear".

1. $f(x) = x + 1$ is called a linear function because its graph is a line
2. $f(x) = x + 1$ is not a linear transformation from a vector space R into R because it preserves neither vector addition nor scalar multiplication

Introduction to Linear Transformations

Zero transformation

$$T : V \longrightarrow W \qquad T(\mathbf{v}) = \mathbf{0}, \qquad \forall \mathbf{v} \in V$$

Identity transformation

$$T : V \longrightarrow W \qquad T(\mathbf{v}) = \mathbf{v}, \qquad \forall \mathbf{v} \in V$$

Theorem ((6.1) Properties of linear transformations)

$$T : V \longrightarrow W, \qquad \forall \mathbf{u}, \mathbf{v} \in V$$

1. $T(\mathbf{v}) = \mathbf{0}$ $(T(cv) = cT(v) \mathit{for} c = 0)$
2. $T(-\mathbf{v}) = -T(\mathbf{v})$ $(T(cv) = cT(v) \mathit{for} c = -1)$
3. $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$ $(T(\mathbf{u} + (-\mathbf{v})) = T(\mathbf{u}) + T(-\mathbf{v})$ *and property (2))*
4. *If* $\mathbf{v} = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$ *then*

$$\begin{aligned} T(\mathbf{v}) =& T(c_1 v_1 + c_2 v_2 + \cdots + c_n v_n) \\ =& c_1 T(v_1) + c_2 T(v_2) + \cdots + c_n T(v_n) \end{aligned}$$

(Iteratively using $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ *and* $T(c\mathbf{v}) = cT(\mathbf{v})$*)*

Example (Linear transformations and bases)

Let $T : R^3 \longrightarrow R^3$ be a linear transformation such that

$$
\begin{aligned}
T(1,0,0) &= (2,-1,4) \\
T(0,1,0) &= (1,5,-2) \\
T(0,0,1) &= (0,3,1)
\end{aligned}
$$

Find $T(2,3,-2)$

Solution

$$(2, 3 - 2) = 2(1, 0, 0) + 3(0, 1, 0) - 2(0, 0, 1)$$

(According to the fourth property on the previous slide that state that)
$T(\mathbf{v}) = T(c_1 v_1 + c_2 v_2 + \cdots + c_n v_n) = c_1 T(v_1) + c_2 T(v_2) + \cdots + c_n T(v_n)$

$$\begin{aligned} T(2, 3, -2) =& 2T(1, 0, 0) + 3T(0, 1, 0) - 2T(0, 0, 1) \\ =& 2T(2, -1, 4) + 3T(1, 5, -2) - 2T(0, 3, 1) \\ =& (7, 7, 0) \end{aligned}$$

Example (A linear transformation defined by a matrix)

The function $T : R^2 \longrightarrow R^3$ is defined as

$$T(\mathbf{v}) = A\mathbf{v}$$
$$= \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

(a) Find $T(\mathbf{v})$, where $\mathbf{v} = (2, -1)$

(b) Show that T is a linear transformation from R^2 into R^3.

Solution

(a) *Given* $\mathbf{v} = (2, -1)$, *then*

$$\begin{aligned} T(\mathbf{v}) =& A\mathbf{v} \\ =& \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix} \end{aligned}$$

$$\therefore\ T(2, -1) = (6, 3, 0)$$

(b) $T(\mathbf{u} + \mathbf{v}) = A(\mathbf{u} + \mathbf{v}) = A(\mathbf{u}) + A(\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ *(Vector addition)*
$T(c\mathbf{u}) = A(c\mathbf{u}) = c(A\mathbf{u}) = cT(\mathbf{u})$ *(scalar multiplication addition)*

Theorem ((6.2) The linear transformation defined by a matrix)

Let A be an $m \times n$. The function T defined by

$$T : V = A\mathbf{v}$$

is a linear transformation from R^n into R^m

Note

$$A\mathbf{v} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11}v_1 & + & a_{12}v_2 & + & \cdots & + & a_{1n}v_n \\ a_{21}v_1 & + & a_{22}v_2 & + & \cdots & + & a_{2n}v_n \\ & & \vdots & + & & & \\ a_{m1}v_1 & + & a_{m2}v_2 & + & \cdots & + & a_{mn}v_n \end{bmatrix}$$

$T(\mathbf{v}) = A(\mathbf{v}) \qquad T : R^n \longrightarrow R^m$

Introduction to Linear Transformations

Note

- If $T(\mathbf{v})$ can represented by $A\mathbf{v}$, then T is a linear transformation
- If the size of A is $m \times n$, then the domain of T is R^n and the codomain of T is R^m

Example (Rotation in the plane)

Show that the linear transformations $T : R^2 \longrightarrow R^2$ given by the matrix

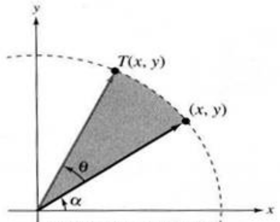
$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

has the property that it rotates every vector in R^2 counterclockwise about the origin through the angle θ

Solution

Let $\mathbf{v} = (x, y) = (r\cos\alpha, r\sin\alpha)$, *where*

- r: *the length of* $\mathbf{v} = \sqrt{x^2 + y^2}$
- α: *the angle from the positive x-axis counterclockwise to the vector* $\mathbf{v}$



Rotation in the Plane

Solution

$$T(\mathbf{v}) = A(\mathbf{v}) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} r\cos\alpha \\ r\sin\alpha \end{bmatrix}$$

$$= \begin{bmatrix} r\cos\theta\cos\alpha & -\sin\theta\sin\alpha \\ r\sin\theta\cos\alpha & r\cos\theta\sin\alpha \end{bmatrix}$$

$$= \begin{bmatrix} r\cos(\theta+\alpha) \\ r\sin(\theta+\alpha) \end{bmatrix}$$

- *r: remain the same, that means the length of $T(\mathbf{v})$ equals the length of $\mathbf{v}$*
- *$\theta + \alpha$: the angle from the positive x-axis counterclockwise to the vector $T(\mathbf{v})$*

Thus, $T(\mathbf{v})$ is the vector that results from rotating the vector $\mathbf{v}$ counterclockwise through the angle θ

Example (A projection in R^3)

The linear transformation $T : R^3 \longrightarrow R^3$ given by the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

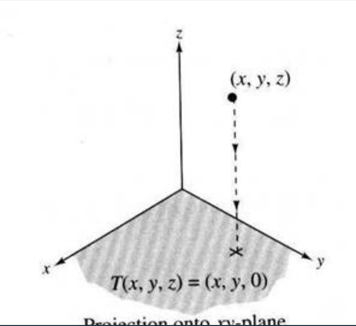
is called a projection in R^3 .

Introduction to Linear Transformations

Solution

If **v** *is* (x, y, z),

$$A(\mathbf{v}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$



Example (The transpose function is a linear transformation from $M_{m\times n}$ into $M_{n\times m}$)

Let the mapping $T : M_{m\times n} \longrightarrow M_{n\times m}$ be defined as

$$T(A) = A^T$$

Show that T is a linear transformation.

Solution

Let $A, B \in M_{m\times n}$*, then*

$$\begin{aligned} T(A+B) =& (A+B)^T = A^T + B^T = T(A) + T(B) \\ T(cA) =& (cA)^T = cA^T = cT(A) \end{aligned}$$

Therefore, The T *(the transpose function) is a linear transformation from* $M_{m\times n}$ *into* $M_{n\times m}$

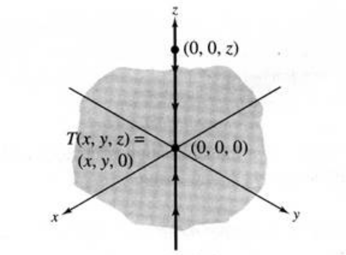
Kernel of a linear transformation T

Let $T : V \longrightarrow W$ be a linear transformation. Then the set of all vectors $\mathbf{v}$ in V that satisfy $T(\mathbf{v}) = \mathbf{0}$is called the kernel of T and is denoted by $\ker(T)$. That is

$$\ker(T) = \{\mathbf{v} | T(\mathbf{v}) = \mathbf{0}, \forall \mathbf{v} \in V\}$$

Example (The transpose function is a linear transformation from $M_{m \times n}$ into $M_{n \times m}$)

- For example, V is R^3, W is R^3, and T is the orthogonal projection of any vector (x, y, z) onto the xy-plane, i.e. $T(x, y, z) = (x, y, 0)$
- Then the kernel of T is the set consisting of $(0, 0, s)$, where s is a real number, i.e
- $\ker(T) = \{(0, 0, s) | s \text{ is a real number}\}$



The kernel of T is the set of all vectors on the z-axis.

Example (Finding the kernel of a linear transformation)

$$T(A) = A^T \qquad (T : M_{3\times 2} \longrightarrow M_{2\times 3})$$

Solution

$$ker(T) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Example (The kernel of the zero and identity transformations)

- If $T(\mathbf{v}) = \mathbf{0}$ (the zero transformation $T : V \longrightarrow W$), then $\ker(T) = V$
- If $T(\mathbf{v}) = \mathbf{v}$ (the identity transformation $T : V \longrightarrow V$), then $\ker(T) = \{\mathbf{0}\}$

The Kernel and Range of a Linear Transformation

Example (Finding the kernel of a linear transformation)

$$T(\mathbf{x}) = A\mathbf{v} = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad (T : R^3 \longrightarrow R^2)$$

ker(T)=?

Solution

$$ker(T) = \{x_1, x_2, x_3) | T(x_1, x_2, x_3) = (0,0), \ \ and \ (x_1, x_2, x_3) \in R^3\}$$

$$T(x_1, x_2, x_3) = (0,0)$$

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 1 & -1 & -2 & 0 \\ -1 & 2 & 3 & 0 \end{bmatrix} \xrightarrow{G.-J.E} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\implies ker(T) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} ker(T) &= \{t(1,-1,1) | t \textit{ is a real number } \} \\ &= span\{(1,-1,1)\} \end{aligned}$$

Theorem ((6.3) The kernel is a subspace of V)

The kernel of a linear transformation $T : V \longrightarrow W$ is a subspace of the domain V

Proof.

$\because$ $T(\mathbf{v}) = \mathbf{0}$ (by Theorem 6.1)
$\therefore$ $\ker(T)$ is a nonempty subset of V
Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the kernel of T. Then,

$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

$(\mathbf{u} \in ker(T), \mathbf{v} \in ker(T) \implies \mathbf{u}+\mathbf{v} \in ker(T))$

$$T(c\mathbf{u}) = cT(\mathbf{u}) = cT(\mathbf{0}) = \mathbf{0}$$

$(\mathbf{u} \in ker(T) \implies c\mathbf{u} \in ker(T))$
Thus, $\ker(T)$ is a subspace of V (according to Theorem 4.5 that a nonempty subset of V is a subspace of V if it is closed under vector addition and scalar multiplication) $\square$

Example (Finding a basis for the kernel)

Let $T : R^5 \longrightarrow R^4$ be defined by $T(\mathbf{x}) = A\mathbf{x}$, where $\mathbf{x}$ is in R^5 and

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix}$$

Find a basis for ker(T) as a subspace of R^5

Solution

To find ker(T) means to find all $\mathbf{x}$ satisfying $T(\mathbf{x}) = A\mathbf{x} = \mathbf{0}$. Thus we need to form the augmented matrix $[A|\mathbf{0}]$ first

The Kernel and Range of a Linear Transformation

Solution

$$[A|\mathbf{0}] =$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & -1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 0 \\ -1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 8 & 0 \end{bmatrix} \xrightarrow{G.-J.E} \begin{bmatrix} 1 & 0 & 2 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2s+t \\ s+2t \\ s \\ -4t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 0 \\ -4 \\ 1 \end{bmatrix}$$

$B = \{(-2, 1, 1, 0), (1, 2, 0, -4, 1)\}$: *one basis for the kernel of T*

Matrices for Linear Transformations

Next week!!!

Transition Matrices and Similarity

Next week!!!